

Pandit Deendayal Energy University

End Semester Examination, May - 2024

B. Tech (Computer Engineering)

Semester - VI

Course Name: Big Data Analytics

Course Code: 23CP309T

Date: 11/05/2024

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Question No.	Description	Marks	Course Outcomes (COs)
Q1-A	For a healthcare use case scenario, where does the volume, velocity, and variety of data necessitate the adoption of big data analytic techniques?	10	CO1
	OR		
Q1-A	Draw and explain Hadoop architecture with eco system.	10	CO1
Q1-B	Answer the following: (Attempt any Two)		
	1. How does data flow occur in HDFS, and what mechanisms ensure fault tolerance and data reliability?	5	CO2
	2. Design a MapReduce algorithm to calculate the total number of occurrences of various words in a dataset using the map, shuffle, combiner, and reduce phases. Assume that the dataset consists of a large size file, and your task is to efficiently count the occurrences of each word across the document.	5	CO2
	3. Describe the role of visualization libraries and tools in creating insightful visualizations for big data analysis.	5	CO2
	4. How is reliability ensured in Hadoop?		
Q.2.	Answer the following: (Attempt any Four)		
	1. Explain the concept of RDD and their role in spark programming.	5	CO4
	2. How does Spark Streaming facilitate the processing of continuous streams of data, and what are its advantages over batch processing frameworks like MapReduce?	5	CO4
	3. Describe the flow of data in distributed file system.	5	CO1
	4. List five transformation functions of Spark with suitable example.	5	CO4
	5. Discuss different persistence strategies in Spark.	5	CO4
Q.3.	Answer the following:		
	1. Compare and contrast Pig Latin with SQL in terms of their syntax, capabilities, and performance for data processing tasks.	5	CO3
	2. Apply the transformation function in Scala to count the occurrences of each word in a text file. Given that file path is stored in variable path_textFile. design appropriate code for the below statements.	5	CO5
	val textFile = _____		
	val counts = _____		

3. Design a Kafka streaming architecture for Uber's ride-booking platform to support real-time data processing and analytics. Discuss the key components, data flow, and use cases within the Kafka ecosystem, considering the challenges and requirements specific to Uber's operations. 10 CO6

Q.4. Answer the following:

1. Design a Pig script to process a large log file containing user activity data. The script should extract relevant information such as user IDs, timestamps, and actions performed, and perform aggregation tasks like counting the number of actions per user, at the end also store the aggregated results. 10 CO3
2. For dataframe df, the df.printSchema() gives the output as shown below: 10 CO3

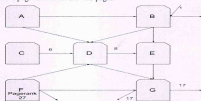
```
 |-- rate_marriage: integer (nullable = true)
 |-- age: double (nullable = true)
 |-- yrs_married: double (nullable = true)
 |-- children: double (nullable = true)
 |-- religious: integer (nullable = true)
 |-- affairs: integer (nullable = true)
```

Develop a Spark SQL query to perform the following:

- The average age of individuals involved in marital affairs.
- The maximum number of years married among individuals with at least one child.
- The percentage of marriages rated above 4 (considering 5 as the maximum rating) among individuals who reported having affairs.
- The average age of individuals involved in affairs, grouped by the number of children they have.

Q.5. Answer the following:

1. Find the page rank of all the web pages. 5 CO5



2. Design a MongoDB Schema for the given RDBMS Schema and perform the following operations on it. 10 CO5
- Design a mongodb command to find product price less than 5000
 - Design a mongodb command to drop the material collection



3. Discuss ML Pipeline in text processing for large-scale datasets with suitable example. 5 CO6